1.1 Mental Health Survey Clustering Analysis ***

1.1.1 Introduction

Mental health is a critical component of overall well-being, yet it remains a topic that many people find challenging to discuss or address. This study aims to uncover patterns in mental health data using clustering techniques, which can help identify groups of individuals with similar mental health profiles. By understanding these patterns, we can better tailor support and interventions to different groups of people.

The dataset used in this analysis contains various features related to mental health, including demographic information, publication details, and specific mental health indicators. The goal is to apply clustering algorithms to this data to see if we can find meaningful groups that may be useful for mental health professionals or researchers.

Key steps in this analysis include:

1. Data preprocessing
2. Exploratory data analysis
3. Feature selection
4. Dimensionality reduction
5. Clustering algorithm implementation
6. Cluster analysis and interpretation

1.1.2 Part 1: Data Preprocessing and Exploratory Data Analysis (EDA) ***

In this section, we will load the dataset, perform initial exploration, and prepare the data for clustering. This includes handling missing values, encoding categorical variables, and scaling numerical features.

1.1.2.1 Loading Required Libraries

```
# Data manipulation and visualization
library(ggplot2)
library(dplyr)
library(tidyr)
library(gridExtra)
library(corrplot)
library(RColorBrewer)
library(scales)

# Machine learning and clustering
library(cluster)
library(factoextra)
library(NbClust)
library(dbscan)
library(mclust)

# Dimensionality reduction
library(Rtsne)
library(umap)

# Data preprocessing
library(caret)
library(fastDummies)

# Set seed for reproducibility
set.seed(123)
```

1.1.2.2 Loading the Dataset

```
# Load the data
data <- read.csv("mental_health_survey.csv")

# Display first 6 rows
head(data)
```

```
  X publication_info A_Bullying W_Bullying Anxiety Depression Depression_severity
1 0       Unknown|NA         NA         NA      NA         NA                NA
2 1       Unknown|NA         NA         NA      NA         NA                NA
3 2       Unknown|NA         NA         NA      NA         NA                NA
4 3       Unknown|NA         NA         NA      NA         NA                NA
5 4       Unknown|NA         NA         NA      NA         NA                NA
6 5       Unknown|NA         NA         NA      NA         NA                NA
  PTSD Psychological_disorder Mental_Health_Treatment Substance_abuse
1   NA                    NA                       NA             NA
2   NA                    NA                       NA             NA
3   NA                    NA                       NA             NA
4   NA                    NA                       NA             NA
5   NA                    NA                       NA             NA
6   NA                    NA                       NA             NA
  Addiction Neuroticism Extraversion Agreeableness Conscientiousness Openness
1        NA         NA          NA           NA               NA       NA
2        NA         NA          NA           NA               NA       NA
3        NA         NA          NA           NA               NA       NA
4        NA         NA          NA           NA               NA       NA
5        NA         NA          NA           NA               NA       NA
6        NA         NA          NA           NA               NA       NA
  Publication
1          NA
2          NA
3          NA
4          NA
5          NA
6          NA
```

```
# Dimensions of the dataset
dim(data)
```

```
[1] 1000   21
```

```
# Column names
colnames(data)
```

```
 [1] "X"                    "publication_info"     "A_Bullying"           "W_Bullying"           "Anxiety"              "Depression"           "Depression_severity"  "PTSD"                 "Psychological_disorder"
 [10] "Mental_Health_Treatment" "Substance_abuse"      "Addiction"            "Neuroticism"          "Extraversion"         "Agreeableness"        "Conscientiousness"    "Openness"             "Publication"
```

```
# Structure of the data
str(data)
```

```
'data.frame':	1000 obs. of  21 variables:
 $ X                    : int  0 1 2 3 4 5 6 7 8 9 ...
 $ publication_info     : chr  "Unknown|NA" "Unknown|NA" "Unknown|NA" "Unknown|NA" ...
 $ A_Bullying           : int  NA NA NA NA NA NA NA NA NA NA ...
 $ W_Bullying           : int  NA NA NA NA NA NA NA NA NA NA ...
 $ Anxiety              : int  NA NA NA NA NA NA NA NA NA NA ...
 $ Depression           : int  NA NA NA NA NA NA NA NA NA NA ...
 $ Depression_severity  : int  NA NA NA NA NA NA NA NA NA NA ...
 $ PTSD                 : int  NA NA NA NA NA NA NA NA NA NA ...
 $ Psychological_disorder: int  NA NA NA NA NA NA NA NA NA NA ...
 $ Mental_Health_Treatment: int  NA NA NA NA NA NA NA NA NA NA ...
 $ Substance_abuse      : int  NA NA NA NA NA NA NA NA NA NA ...
 $ Addiction            : int  NA NA NA NA NA NA NA NA NA NA ...
 $ Neuroticism          : int  NA NA NA NA NA NA NA NA NA NA ...
 $ Extraversion         : int  NA NA NA NA NA NA NA NA NA NA ...
 $ Agreeableness        : int  NA NA NA NA NA NA NA NA NA NA ...
 $ Conscientiousness    : int  NA NA NA NA NA NA NA NA NA NA ...
 $ Openness             : int  NA NA NA NA NA NA NA NA NA NA ...
 $ Publication          : int  NA NA NA NA NA NA NA NA NA NA ...
```

1.1.2.3 Initial Data Exploration

```
# Summary statistics
summary(data)
```

```
       X      publication_info   A_Bullying      W_Bullying       Anxiety         Depression      Depression_severity      PTSD           Psychological_disorder Mental_Health_Treatment Substance_abuse
 Min.   :   0   Length:1000        Min.   :1.0   Min.   :1.0   Min.   :1.0   
```